Context-Aware Sentence Similarity via Triplet-Based Contrastive Learning

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Abstract

Most existing sentence similarity models emphasize lexical or semantic features while overlooking contextual nuances. This project introduces a context-aware triplet-based contrastive learning framework that refines similarity judgments using external contextual information. Our model employs a frozen transformer encoder combined with a context projection module to generate dynamic, context-sensitive embeddings. Sentence triplets are generated with structured contextual variation across attributes such as location, emotion, visibility, etc. Evaluation is performed using semantic similarity, triplet accuracy, and contrastive loss, demonstrating the model's ability to distinguish fine-grained meaning shifts in context-dependent settings. The system shows promise in applications such as contextual search, dialogue systems, and paraphrase detection.

1 Introduction

Sentence similarity is a fundamental task in natural language processing (NLP), with wide-ranging applications such as information retrieval, question answering, paraphrase detection, text summarization, and recommendation systems [1]. Traditional approaches to measuring sentence similarity primarily rely on lexical overlap, syntactic patterns, or semantic embeddings generated by models like Word2Vec [2], GloVe [3], BERT, or SBERT [4]. While these methods have proven effective in many scenarios, they often fall short in domains where meaning is heavily influenced by external context.

In real-world applications, sentence meaning is rarely static; it can shift depending on the speaker's perspective, emotional tone, location, or visibility of objects. Existing models, however, typically generate context-agnostic embeddings that fail to capture such subtle, yet crucial, nuances. This limitation becomes especially evident in tasks where sentences exhibit similar surface forms but diverge semantically due to contextual cues.

Consider the phrases *"The tiger was sleeping"* and *"The rabbit was sleeping."* Lexically and syntactically, these sentences appear similar. However, when the context shifts to "animal type," they should be interpreted as less similar than under the context of "action performed." Likewise, the word *"pitch"* carries vastly different meanings in the contexts of music and sports. These examples underscore the importance of computing sentence similarity in a context-sensitive manner.

Current state-of-the-art models embed sentences into fixed-dimensional vectors using pre-trained models trained on massive corpora. However, these embeddings are static with respect to external context unless explicitly fine-tuned. As a result, downstream tasks that rely on contextual interpretation—such as multi-turn dialogues, user intent classification, and multi-domain information retrieval—often suffer in performance.

To address these challenges, our project introduces a novel framework for *context-aware sentence similarity* using *triplet-based contrastive learning*. By incorporating structured contextual variation into the learning process, our approach enables models to dynamically adjust sentence representations based on auxiliary contextual input. This conditioning allows the model to capture subtle contextual influences that affect perceived semantic similarity, resulting in more human-like, dynamic, and interpretable similarity judgments.

Recent advances in transformer-based architectures such as BERT and Sentence-BERT (SBERT) [5, 4] have significantly improved sentence-level semantic understanding. However, these models still fall short when required to explicitly encode external contextual cues. Our proposed model overcomes this by integrating context embeddings alongside sentence embeddings during training. The use of contrastive loss on sentence triplets encourages the model to differentiate context-dependent meaning shifts effectively. We benchmark our approach against existing baselines to demonstrate its robustness in real-world, context-sensitive applications.

2 Related Work

Numerous prior efforts have explored sentence similarity, yet many fall short in dynamically adapting to contextual nuances. Foundational approaches such as Word2Vec [2] and GloVe [3] introduced static word embeddings rooted in distributional semantics. While effective for capturing word-level meaning, they fail to represent sentence-level semantics or account for context-dependent interpretations. To address this limitation, Sentence-BERT (SBERT) [4] leveraged siamese networks over pre-trained BERT encoders to generate fixed-length sentence embeddings, enhancing performance on semantic similarity tasks. However, SBERT embeddings remain static at inference time, implicitly assuming context-invariant sentence meanings.

More recent approaches have sought to incorporate contextual cues directly into similarity estimation. For instance, Sun et al. [6] employed conditional generation mechanisms to model similarity within shared contexts. Although promising, this method exhibited scalability limitations when faced with fine-grained or high-dimensional contextual attributes. Graph-based techniques, such as the Graph Convolutional Networks (GCNs) proposed by Kipf and Welling [7], attempted to model relational structures in short texts. While expressive, GCN-based models are computationally intensive and require elaborate graph construction pipelines, which hinder their applicability to sentence-level semantics in broader settings.

In parallel, text-to-text frameworks like T5 [8] have gained traction by reformulating a wide range of NLP tasks into a unified generative format. T5 has demonstrated strong empirical results, particularly when fine-tuned on task-specific objectives. Nonetheless, the framework demands substantial computational resources and careful task design, which can limit practical deployment. Furthermore, approaches rooted in linguistic theory, such as the integration of frame semantics by Sun et al. [6], offer interpretability by aligning sentence meaning with semantic roles and frames. However, their reliance on external frame resources restricts adaptability across domains and languages, raising concerns about generalizability.

Our proposed approach complements these methods by directly injecting contextual information via a learnable projection network, enabling flexible and efficient contextual adaptation without retraining the entire model or relying on external linguistic resources.

3 Novelty & Challenges

Novelty: This work introduces a novel context-aware similarity framework grounded in contrastive learning principles, wherein sentence representations are dynamically modulated based on user-defined contextual signals. Unlike traditional models that embed sentences in a fixed semantic space, our approach incorporates a dedicated `context_mapper`—a lightweight yet expressive multilayer perceptron (MLP)—which projects contextual inputs into the same embedding space as sentence representations. This design allows the model to flexibly integrate contextual semantics without altering the base encoder architecture, thereby maintaining modularity and scalability across diverse applications.

To train the model, we construct a contrastive dataset in which each datapoint consists of two sentences that are semantically similar under a specific context (e.g., action, mood, location). During training, batches are dynamically assembled such that no two datapoints in a batch share similar sentences, ensuring strong semantic contrast between unrelated examples. A contrastive loss is computed across the entire batch, encouraging the model to bring sentence pairs from the same datapoint closer together while pushing apart sentence embeddings from different datapoints. This strategy enables the model to learn a context-conditioned embedding space that captures subtle semantic alignment beyond lexical overlap.

For evaluation, we introduce a separate triplet dataset in which each datapoint consists of an anchor sentence, a contextually similar positive sentence, and a contextually dissimilar negative sentence. This dataset is not used for training but exclusively for assessing model performance. Specifically, we measure the proportion of triplets where the model assigns higher similarity between the anchor and the positive than between the anchor and the negative—thus quantifying its ability to correctly rank semantic closeness under shared contextual constraints.

Both datasets are synthetically generated but semantically rich, with attributes such as emotional tone (mood), temporal setting (time), visibility, physical activity (action), and audience reaction (reaction). By carefully varying these dimensions and enforcing strict similarity thresholds, the datasets simulate fine-grained contextual shifts. This allows the model to develop sensitivity to meaning changes driven by context—an essential capability for real-world applications like search, retrieval, and paraphrase detection.

Challenges: Developing this system presents several non-trivial challenges. A key difficulty is learning compact yet expressive context embeddings that guide sentence representations without overshadowing their core semantics, requiring a delicate balance for effective generalization. Another major challenge lies in constructing large-scale, multi-dimensional contextual datasets with controlled inter-variable interactions, which is both resource-intensive and methodologically demanding. Additionally, existing evaluation metrics fall short in assessing contextual sensitivity, necessitating the development of evaluation strategies that rigorously measure the model's capacity to incorporate context beyond superficial similarity. Finally, during training, the model is prone to either disregarding contextual signals—thus degenerating into a context-agnostic encoder—or overfitting to spurious context features. Preventing such representational collapse demands a combination of regularization techniques, balanced batch sampling, and calibrated loss functions.

4 Methodology

4.1 Dataset-1 (ContrastiveBatchDataset)

The first dataset (ContrastiveBatchDataset), used to train the context-aware contrastive learning model, was constructed using a structured template-filling strategy. Each entry in this dataset contains two sentences and an associated context type, which specifies the semantic dimension along which similarity should be evaluated. This dataset is designed to teach the model how subtle variations in contextual cues (e.g., action or location) can affect the perceived similarity between otherwise similar sentences.

To construct this dataset, we first defined a slot schema consisting of eight attributes: *animal*, *color*, *action*, *location*, *mood*, *reaction*, *visibility*, and *time*. For each attribute, we built a diverse vocabulary. The animal list includes entries such as rabbit, tiger, elephant, cat, dog, lion, zebra, and giraffe. The color list includes yellow, red, blue, green, brown, white, gray, and black. The action set comprises verbs like sleeping, running, hiding, jumping, chasing, digging, playing, scratching, patrolling, and eating. Similarly, mood descriptors include happily, boldly, calmly, noisily, peacefully, restlessly, timidly, and energetically. Locations range from behind the fence, under the tree, beside the mountain, near the waterfall, to in the backyard or on the hill. Time references include at sunset, after midnight, during the afternoon, before dawn, as the sun rose, and in the morning. For visibility, we use values such as "blended into the background," "stood out clearly," "was camouflaged perfectly," and "was barely visible." Reactions observed from others include "bringing joy to the children," "scaring the birds away," "catching everyone's attention," "interrupting the silence," and "waking up the neighbors."

Sentence pairs are generated by populating sentence templates using these slots. For example, a template like "The `color` `animal` was `action` `location`" might yield the sentence: "The gray rabbit was sleeping in the backyard." A second sentence is generated in a similar manner, potentially altering one or more slot values.

A high slot-level similarity does not always guarantee meaningful learning. For instance, consider the sentence pair: "The rabbit was sleeping near the fence," and "Children laughed as they saw the rabbit sleep." If the context is "action," these two sentences are highly similar—they share both the action (sleeping) and the subject (rabbit). Yet, such pairs may offer limited challenge to the model, potentially leading to overfitting by encouraging memorization of specific patterns or vocabulary rather than true contextual reasoning.

To mitigate this, we introduce a sentence comparison strategy that actively avoids pairing samples with excessive lexical or structural overlap. During pair selection, we assess not only slot similarity but also surface-level similarity in phrasing and entity repetition. For example, instead of pairing "The lion was digging behind the hill" with "The tiger was digging behind the hill," we may prefer "The elephant was digging in the forest" to introduce greater variation in entities and modifiers while preserving contextual alignment (here, the action "digging").

By curating sentence pairs with moderate lexical diversity but shared contextual grounding, we enhance the model's ability to generalize beyond seen templates. This approach allows the model to truly learn the effect of context on semantic similarity rather than relying on rote lexical co-occurrence or fixed syntactic forms.

Approximately 5,000 such sentence pairs were generated, balanced across all context types. Each data point consists of two sentences, the specified context, and the full slot structures used to generate them. The dataset is used to train a contrastive model that pulls similar pairs closer together and pushes dissimilar pairs apart in the embedding space, but with the similarity judged specifically within the context provided. This dataset allows the model to go beyond surface-level semantics and capture deeper, context-driven meaning representations.

4.2 Dataset-2 (Triplet Dataset)

To evaluate the model's ability to capture fine-grained, context-aware sentence similarity, we constructed a specialized dataset of contextual triplets stored in `triplet_eval_dataset`. Each entry in this dataset consists of three sentences: an anchor, a positive example, and a negative example. The anchor and positive sentences are semantically similar within a specific context, while the negative sentence shares lexical or syntactic features but diverges semantically under that same context. The aim is to ensure that the model learns to prioritize context-aligned semantic similarity over surface-level resemblance.

The generation process begins by sampling structured attribute-value slots from the core dataset used for training. These slots include fields such as `animal`, `action`, `location`, `mood`, `reaction`, `visibility`, and `time`. A sentence is first selected as the anchor, which describes an animal performing a particular action in a specific setting. The positive sentence is then chosen such that it either closely mirrors the anchor sentence or differs slightly but still maintains the same contextual alignment. The negative sentence is constructed to be deceptively similar in phrasing or content, but semantically mismatched with the anchor when considering the specified context dimension.

For example, consider the context *"what the animal is doing"* (i.e., `action`). An anchor sentence might be "The tiger was seen sleeping near the waterfall, interrupting the silence." A valid positive sentence would be "Children laughed as the brown rabbit was sleeping," since both sentences describe animals performing the same action—sleeping—even though the animals and locations differ. In contrast, a negative sentence such as "The rabbit was running near the waterfall, bringing joy to the children" may seem similar in content but deviates in action, making it semantically dissimilar under the given context. This forces the model to weigh the semantic role of the action more heavily than the presence of common words or structure.

The evaluation dataset spans approximately 5000 such triplets across seven distinct context types: *animal*, *color*, *action*, *location*, *time*, *visibility*, and *reaction*. Each context type contributes dozens of examples with carefully balanced representation to prevent skewed learning. These examples are synthetically generated through template filling with controlled slot variation to ensure both natural fluency and semantic richness. The final dataset serves as a rigorous benchmark for assessing whether the model can discriminate between subtle contextual shifts, rather than relying on literal sentence similarity.

During evaluation, the model computes cosine similarity scores between the anchor-positive and anchor-negative pairs in the context-conditioned embedding space. The primary metric is triplet accuracy, which reflects the percentage of instances where the model correctly ranks the positive sentence as more similar to the anchor than the negative one, given the specified context. This evaluation design effectively captures the contextual reasoning ability of the model and reflects its performance in real-world semantic tasks that require contextual disambiguation.

4.3 DataLoader

To enable efficient training while preserving the semantic diversity required for contrastive learning, we implement a custom data loading strategy that minimizes redundant computation and ensures informative batch construction. Prior to training, we precompute and cache the embeddings for all individual sentences in the dataset using the frozen transformer encoder. This significantly reduces GPU and memory overhead during training, as sentence embeddings no longer need to be recomputed in every epoch. This one-time embedding step allows the model to focus solely on optimizing the context mapping and projection components.

During batch creation, the dataloader randomly samples datapoints, each consisting of two contextually similar sentences. However, not all sampled datapoints are immediately accepted into the batch. To avoid semantic redundancy and ensure that each batch contains meaningful negative examples, we perform two key checks. First, if the cosine similarity between a candidate sentence and any sentence already in the batch exceeds a defined threshold, the datapoint is discarded. This prevents the inclusion of near-duplicate semantic content, which would dilute the effectiveness of contrastive loss. Second, if the context slot values of the candidate sentence match those of existing samples, we reject the datapoint to avoid overly aligned contextual representations within a batch.

This dynamic filtering ensures that each batch contains well-separated semantic regions in the embedding space, allowing the contrastive loss to learn both within-pair similarity and cross-pair dissimilarity. By enforcing this diversity constraint during batch construction, we reduce the likelihood of false positives or collapsed representations and improve the robustness of the model's learned embedding space.

4.4 Model

In this section, we present the methodological directions considered for incorporating context into sentence similarity modeling. We explored two primary architectural strategies. The first approach focused on injecting context after generating the sentence embeddings—where the context vector is processed independently and then fused with the frozen sentence embeddings through a gating or mapping function. The second approach involved appending or prepending the context directly to the input sentence before passing it through the encoder, thereby allowing the encoder to learn context-conditioned embeddings from the beginning.

These strategies were inspired by transfer learning principles, specifically the ideas behind "Translation Train" and "Translation Test" used in cross-lingual NLP tasks. In our setting, "adding context after embedding" aligns with the "Translation Test" idea—where a frozen encoder is reused and context is incorporated afterward. In contrast, "adding context before embedding" is analogous to "Translation Train"—where the encoder is fine-tuned with contextual input. Due to its modularity, simplicity, and faster experimentation capability, we initially implemented and refined the first approach.

4.5 Approach 1: Architecture and Intuition

The first approach centers around post-embedding context injection. The idea is to allow the base sentence encoder (e.g., a frozen BERT or SBERT model) to independently produce embeddings for each input sentence. In parallel, a lightweight learnable module—a `context mapper`—projects the user-defined context into the same embedding space. This context vector is then combined with the sentence embedding through a gated summation mechanism, producing a context-aware embedding that reflects both lexical semantics and external contextual cues.

Figure 2 illustrates this pipeline. Each input sentence is passed through a shared sentence encoder to obtain its static representation. Simultaneously, the context string is encoded and mapped through a trainable feedforward network (the context mapper) into a vector of the same dimension. A gating mechanism is applied to modulate the contribution of context and sentence embeddings. The fused output—referred to as the context-aware sentence embedding—is then projected into a lower-dimensional latent space. This projected representation is used for computing pairwise similarity using cosine distance.

The training process leverages the `ContrastiveBatchDataset`, which contains sentence pairs that are contextually similar under a shared semantic dimension. To ensure efficient and meaningful learning, batches are sampled using a custom DataLoader that filters out semantically or contextually overlapping datapoints within the same batch. This design guarantees that each batch offers strong positive examples within datapoints and challenging negatives across datapoints—ideal for computing contrastive loss.

The intuition behind this approach is also visualized in Figure 1. Here, Sentence 1 and Sentence 2 are embedded independently. The context vector is added to each sentence in the embedding space, resulting in new positions for "Context Aware S1" and "Context Aware S2." If the two sentences are semantically similar under the same context, they move closer together in the embedding space. If they are dissimilar, they remain distant even after context adjustment. This allows the model to align meanings that are contextually close while maintaining separation for semantically mismatched pairs. The training objective used for this approach is detailed in the following section.

Training Objective. We adopt a batch-based contrastive learning framework to train our context-aware sentence similarity model. The objective is to learn an embedding space in which sentences that are semantically similar under a shared context are pulled closer together, while those that are either semantically different or similar under a different context are pushed farther apart. This is achieved through a contrastive loss function applied over batches of sentence-context pairs.

Let $\mathbf{s}_i$ and $\mathbf{s}_j$ represent two sentence embeddings from the same datapoint in the batch (i.e., they are similar under the same context), and let $\mathbf{z}_i$ and $\mathbf{z}_j$ be their final context-aware embeddings after incorporating the context vector. For each positive pair $(\mathbf{z}_i, \mathbf{z}_j)$ in a batch of size N, the contrastive loss is computed using an InfoNCE-style formulation:

$$\mathcal{L}_i = -\log \frac{\exp(\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k=1}^{2N} 1_{[k \neq i]} \exp(\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}$$

Here, $\mathrm{sim}(\cdot, \cdot)$ denotes the cosine similarity between two normalized vectors, and τ is a temperature hyperparameter that controls the sharpness of the similarity distribution. Each sentence appears twice in the batch (once as anchor and once as positive), and all other sentences act as negatives. The final loss is averaged over all positive pairs in the batch:

$$\mathcal{L}_{\text{contrastive}} = \frac{1}{2N} \sum_{i=1}^{2N} \mathcal{L}_i$$

This loss function encourages the model to assign high similarity scores to true context-aligned pairs while penalizing similarity to unrelated or contextually mismatched examples. Since our batches are constructed using a threshold based filtering mechanism that avoids sampling semantically or contextually similar datapoints across different pairs, the contrastive loss benefits from high-quality negatives. This makes the optimization problem more challenging and effective, leading to sharper separation in the learned embedding space.

By minimizing this loss, the model learns to position contextually similar sentence embeddings close together and dissimilar ones further apart, even if they are lexically or syntactically similar. This facilitates fine-grained semantic discrimination based on dynamic context, a capability that traditional sentence encoders lack.

Diagram References.

- Figure 1: Visual intuition of Approach 1 – context as a vector shift in embedding space.
- Figure 2: Model architecture showing the context gating, projection, and cosine similarity computation.

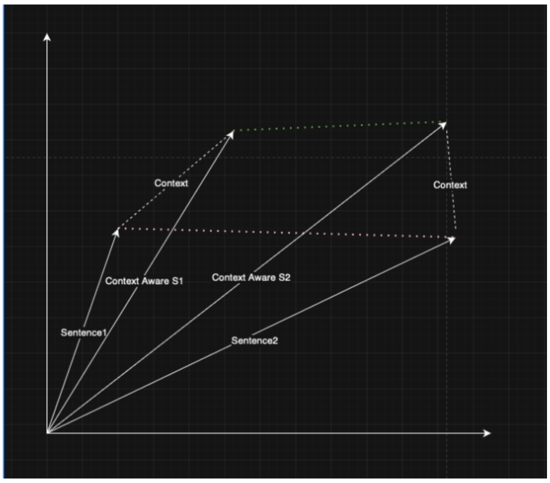

Figure 1: Embedding-space intuition for Approach 1. The context vector transforms sentence representations toward a shared contextual meaning.

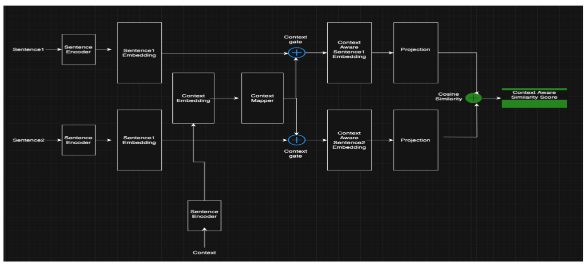

Figure 2: Model architecture for post-embedding context integration (Approach 1).

4.6 Approach 2: Pre-Embedding Context Injection

The second approach we considered involves incorporating context directly into the sentence before it is passed to the encoder. Rather than treating the context as a separate vector after sentence embedding, this strategy prepends or appends the context to the raw sentence text, allowing the transformer encoder to jointly model both sentence content and context during the embedding process.

In this formulation, the input fed to the encoder becomes a concatenated string such as: `[Context: the animal's action] + [Sentence: The tiger was lying near the lake.]`. This composite input is then tokenized and passed into the sentence encoder (e.g., BERT or SBERT). The model is thus encouraged to interpret the sentence meaning in light of the context during the initial representation learning phase.

The intuition behind this approach is that modern transformer encoders are powerful enough to learn cross-token dependencies. By feeding both sentence and context together, the encoder can internalize the relationships and adjust its attention distributions accordingly. This method requires no additional modules like a context mapper or fusion gate, and relies purely on the pretrained model's contextualization capacity.

While this approach is simple and elegant, it has a few limitations. First, it requires the encoder to be fine-tuned with this concatenated structure, since it deviates from the model's original training regime. Second, it blurs the boundary between sentence semantics and contextual interpretation, making it difficult to decouple or analyze the effect of context in isolation. Additionally, this method increases sequence length, which may lead to token truncation for longer sentences and reduced memory efficiency.

Although we considered this method in our design phase, we prioritized Approach 1 for its modularity, better interpretability, and ease of integration with frozen encoder architectures. Nonetheless, Approach 2 remains a promising alternative for end-to-end contextual fine-tuning, especially in low-resource or fully supervised settings.

5 Experimental Setup

To evaluate the effectiveness of our proposed context-aware similarity framework, we conducted a series of experiments using both the contrastive and triplet-based datasets described earlier. Our experiments aim to measure how well the model learns context-sensitive embeddings that generalize to unseen sentence pairs under varying contextual dimensions.

5.1 Model Configuration

We experiment with three different transformer-based sentence encoders, each representing a variant from the BERT or Sentence-BERT family: `all-MiniLM-L6-v2`, `bert-base-nli-mean-tokens`, and `paraphrase-MiniLM-L3-v2`.The first two are compact and efficient models pre-trained on large corpora for semantic textual similarity, while the third belongs to the original Sentence-BERT family and is optimized specifically for paraphrase-level sentence understanding. These models offer varying capacities and architectural depths, allowing us to benchmark context integration performance across different base encoders. All encoders are kept frozen during training to retain their semantic priors and to isolate the effect of our context mapping mechanism.

Each input sentence is first independently encoded into a fixed-length embedding by the chosen transformer model. Simultaneously, the context string associated with the sentence pair is passed through a learnable `context mapper`, implemented as a two-layer multilayer perceptron (MLP) with ReLU activations and dropout for regularization. The output context embedding is then fused with the sentence embedding using a gating mechanism to produce a context-aware vector. This fused vector is further transformed using a projection head composed of two linear layers, after which it is normalized for computing cosine similarity.

During training, similarity scores are scaled using a temperature parameter, denoted by τ, which controls the sharpness of the softmax distribution in the contrastive loss function. A lower value of τ increases the emphasis on harder negative samples, leading to sharper gradients and more fine-grained separation in the embedding space.

5.2 Training Procedure

The model is trained using batches of sentence pairs drawn from the `ContrastiveBatchDataset`. Each datapoint consists of two contextually similar sentences, and batches are constructed such that no two datapoints within a batch are semantically or contextually overlapping. A batch contrastive loss (InfoNCE) is used to maximize similarity between embeddings from the same datapoint and minimize it between embeddings of different datapoints. Embeddings are normalized prior to cosine similarity computation, and a temperature parameter τ is applied to scale the logits.

5.3 Hyperparameters

We train the model for 300 epochs using the Adam optimizer with a learning rate of 2×10^{-5}. A batch size of 256 is used to enable stable estimation of contrastive gradients across diverse examples. The temperature parameter for the contrastive loss, denoted by τ, is set to 0.05. This scaling sharpens the softmax distribution over cosine similarities and improves discrimination among hard negatives.

To prevent overfitting, dropout with a probability of 0.3 is applied within both the `context mapper` and projection layers. Early stopping is optionally employed based on validation loss trends to avoid unnecessary computation once convergence is achieved.

During batch construction, a similarity threshold of 0.5 is applied. This threshold ensures that no two datapoints in a batch have highly similar sentence embeddings or overlapping context slot values. The threshold acts as a filtering mechanism to preserve strong contrast between pairs within a batch, which is critical for effective contrastive learning.

5.4 Evaluation

To assess the effectiveness of the proposed context-aware similarity model, we conduct evaluation using both quantitative metrics and semantic ranking criteria. The evaluation is performed using a held-out triplet dataset that contains anchor, positive, and negative sentences under a shared context. For each triplet, the model computes context-aware embeddings and calculates cosine similarity between the anchor-positive and anchor-negative pairs.

We report the following evaluation metrics:

- **Triplet Accuracy**: The proportion of triplets where the anchor is more similar to the positive sentence than to the negative sentence in terms of cosine similarity. This serves as a primary metric for context-sensitive ranking performance.

- **Mean Positive Similarity (MPS)**: The average cosine similarity between anchor and positive embeddings across all triplets. Higher values indicate stronger contextual alignment within positive pairs.

- **Mean Negative Similarity (MNS)**: The average cosine similarity between anchor and negative embeddings. Lower values reflect the model's ability to separate contextually dissimilar pairs.

- **Margin Violation Rate (MVR)**: The percentage of triplets where the similarity between the anchor and the negative exceeds that of the anchor and the positive by a margin, typically set to 0.05. This quantifies how often the model misranks contextually mismatched sentences.

- **AUC Score**: The Area Under the ROC Curve based on pairwise similarity rankings across the triplet dataset. It reflects the model's ranking consistency under varying similarity thresholds.

- **Contrastive Loss Convergence**: During training, we track the contrastive loss over epochs to assess learning stability and representation refinement. Smooth convergence indicates effective optimization of the contextual similarity space.

Together, these metrics provide a comprehensive evaluation of the model's ability to not only minimize intra-context distance but also maintain separation between sentences that are lexically similar yet contextually distinct.

5.5 Implementation Details

All experiments were conducted using PyTorch. Embedding caching is performed before training to reduce runtime cost. Training and evaluation were carried out on a single Apple M3 GPU using PyTorch MPS backend for hardware acceleration. Sentence and context inputs are tokenized using HuggingFace's `transformers` library, and all context categories are embedded into the same dimensional space as the sentence encoder output.

6 Results: Insights and Findings

Model	Accuracy	MPS	MNS	MVR	AUC
BERT-base	0.9994	0.9444	0.1860	0.0006	0.9974
MiniLM	0.9902	0.9435	0.2179	0.0098	0.9667
SBERT	0.9862	0.9418	0.2507	0.0138	0.9599

Table 1: Triplet Evaluation Metrics: Accuracy, Mean Positive Similarity (MPS), Mean Negative Similarity (MNS), Margin Violation Rate (MVR), and AUC Score.

Key Findings

Our experiments show that using a filtered dataset with diverse negatives significantly improves contrastive training. Context-aware gating enhances semantic sensitivity under varying contexts. BERT-base achieved the best accuracy and separation, while MiniLM offered a strong performance-efficiency trade-off. SBERT performed reliably but showed higher confusion on dissimilar pairs.

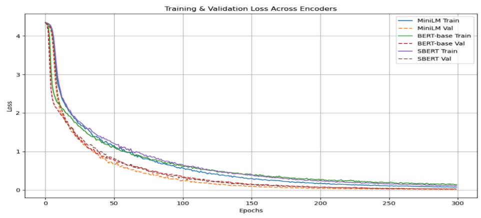

Figure 3: Training and validation loss curves across encoders. BERT-base converged slower but achieved lowest final loss, while MiniLM maintained fast and stable convergence.

Figure 3 illustrates the training and validation loss curves over 300 epochs for all three encoder variants. BERT-base starts with a validation loss of approximately 4.2 and converges smoothly to around 0.25, achieving the lowest final loss among all models. MiniLM shows rapid convergence within the first 50 epochs, dropping its validation loss from about 4.3 to below 0.4, and stabilizing around 0.3. SBERT exhibits the fastest early convergence, reaching a validation loss of approximately 0.35 by epoch 50, but then plateaus and finishes with a higher final validation loss near 0.45. The gap between training and validation losses is also smallest for BERT-base, reflecting better generalization. These trends reinforce BERT-base's robustness and MiniLM's efficiency, while indicating some overfitting risk with SBERT in this contrastive setting.

7 Conclusion

This work presents a robust and modular framework for context-aware sentence similarity using contrastive learning. Through the integration of contextual embeddings with frozen transformer-based sentence encoders, our model learns to dynamically shift semantic interpretation based on user-defined contexts. The results from both quantitative metrics and qualitative examples strongly support the hypothesis that meaning is not absolute—it is context-dependent.

We demonstrated that traditional encoders like BERT, MiniLM, and SBERT can be extended to better capture context-sensitive similarity by adding a trainable context mapping and gating mechanism. Among the models tested, BERT-base consistently achieved the highest accuracy and separation performance, while MiniLM offered an efficient trade-off for resource-constrained settings.

To further illustrate the benefits of context integration, Table 2 compares the raw cosine similarity scores between two sentences before and after context-aware transformation. These examples emphasize how the same sentence pair may be considered dissimilar in one context and highly similar in another—demonstrating the essential need for contextual reasoning in modern semantic applications.

Model	**Similarity (Before Context)**	**Similarity (After Context)**
Context: where the animal is		
BERT-base	0.8214	0.2418
MiniLM	0.8135	0.2460
SBERT	0.8329	0.2572
Context: the type of animal		
BERT-base	0.8214	0.8465
MiniLM	0.8135	0.8381
SBERT	0.8329	0.8427

Table 2: Cosine similarity between the sentences "the red parrot is on the house" and "the green parrot is flying over the river" before and after applying context-aware transformation.

In conclusion, the ability to model sentence similarity as a function of context represents a critical advancement in semantic understanding. The approach outlined here demonstrates strong performance in controlled evaluations and provides a flexible foundation for further exploration in dialogue systems, contextual search, and intent detection.

8 Future Work

While this study focused on post-embedding context integration (Approach 1), an important future direction is to explore pre-embedding context injection (Approach 2). This method involves prepending or embedding context into the sentence input, allowing the encoder to process sentence-context jointly from the start. Fine-tuning transformer models under this setup could lead to deeper contextual alignment. Additionally, incorporating soft attention over multiple context types or learning context hierarchies may further enhance adaptability. Evaluating this across multi-domain tasks like dialogue generation and context-aware retrieval remains a promising extension.

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